

MAHADEVAN REJI

Full-Stack & Systems Developer

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SKILLS

- **Languages:** TypeScript, JavaScript, Go, Rust, Python
- **Systems & Backend:** Node.js, Bun, Go (Fiber), Rust (Axum), Fastify, Hono, FastAPI
- **Infrastructure & Databases:** Cloudflare Workers, MongoDB, PostgreSQL, Redis
- **Core Engineering:** B+ Trees, Storage Engines, Slotted Page Architecture, Vector Search, AOT Compilation

WORK EXPERIENCE

Web and Mobile App Developer Intern

June 2024 - Jul 2024

ZenturioTech | Technopark, Thiruvananthapuram

- Engineered on-device inference capabilities within a React Native application by integrating machine learning models via TensorFlow.js, reducing network dependency and latency.
- Executed the end-to-end development of "Kids Art Studio," an educational app featuring automated drawing analysis and real-time feedback loops.

EDUCATION

B.Tech in Computer Science | CGPA - 8.5

Sept 2023 - Apr 2027

College of Engineering Chengannur, Chengannur, Kerala

PROJECT EXPERIENCE

Bruno

- Built an AOT-compiled radix tree router translating route definitions into optimized if-else chains.
- Achieved peak throughput of 300,000 req/s, optimizing for latency-critical edge runtimes.
- Engineered a custom build-time minification pipeline to improve cold-start performance.

Stack: Typescript, Radix Tree, AOT Compilation, Edge Workers

NanoDB

- Developed a custom storage engine in Go using slotted page architecture for space efficiency.
- Implemented B+ Tree indexing and a native vector search module for similarity-based queries.

Stack: Go, B+ Trees, Slotted Page Architecture, Vector Search

BunkMate

- Engineered local-first architecture for 180+ users with centralized backend synchronization.
- Led full product lifecycle from ideation to deployment based on direct user feedback.

Stack: React Native, Expo, REST API

SnapBook

- Implemented real-time synchronization using WebSockets and event-driven architecture.
- Enabled seamless multi-user collaboration for platform users.

Stack: MongoDB, Express, React Native, Node.js, Websockets